TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Early Detection Technology for Neuro-Musculoskeletal Dysfunction Using Multi-Sensor-Based Smart Shoes for Gait and Balance Data Collection and AI-Based Analysis**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio·Medical		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	*Remote and Home Health Care Devices*	*Embedded Software*	*Intelligent Interpretation Systems*
Concept of Proposed Technology	**Core Concept** This technology is an AI-based digital healthcare solution that continuously collects users' daily gait data through multimodal sensors embedded in smart shoes (including pressure, acceleration, angle, stride length, and speed). By linking these data with clinically validated gait assessment standards such as the TUG Test and SPPB, it enables early detection of neurological (e.g., Parkinson's disease, Alzheimer's disease) and musculoskeletal functional decline, as well as fall risk. **Key Functions and Potential Applications** The collected data are not treated as simple gait measurements, but are transformed into core biomechanical indicators, including physical function assessment metrics, left-right asymmetry, center of pressure (COP) trajectory patterns, temporal variability, and the stability of repetitive gait			

	cycles. By analyzing these indicators based on individualized user profiles, the system is designed to detect abnormal signs at the pre-clinical stage, prior to the onset of overt symptoms, and ultimately to support early disease diagnosis. Furthermore, the technology aims to go beyond conventional data monitoring by providing early risk prediction and functional decline alerts, thereby enabling proactive healthcare interventions. Through continuous and non-invasive monitoring, it supports long-term health management in various environments, including private homes, nursing care facilities, hospitals, and rehabilitation centers. In addition, the proposed system can be integrated with clinical decision-support platforms and digital health services, allowing healthcare professionals to remotely assess patients' mobility status and progression of functional impairment. This facilitates personalized rehabilitation planning, preventive care strategies, and data-driven clinical management, ultimately contributing to improved quality of life and reduced healthcare costs.
Rationale for Joint R&D Project	**Critical Technical Barriers** This technology is an advanced convergent system that organically integrates sensor hardware embedded in smart shoes, clinical gait assessment indicators (TUG and SPPB), artificial intelligence algorithms, and data analysis and management platforms. Due to its highly interdisciplinary nature, encompassing medicine, engineering, artificial intelligence, and data science, comprehensive development and validation across the entire technology lifecycle cannot be sufficiently achieved by the technical, human, and infrastructural capacities of a single institution alone. Moreover, long-term research that simultaneously considers both clinical environments and daily living conditions is essential for ensuring practical applicability and reliability. In addition, technology requires large-scale data accumulation to secure clinical credibility, repeated validation across diverse user groups, and continuous technological advancement to enhance global market competitiveness. Meeting these complex requirements solely within the domestic research environment is constrained by technical and operational limitations, making collaboration with overseas institutions that possess advanced research infrastructure and extensive clinical and field-testing experience indispensable. Furthermore, to achieve successful commercialization and global competitiveness, joint research with international research institutes and industrial partners is crucial for improving technological maturity and verifying applicability across diverse operational environments in advance. Accordingly, this project needs to transition from an independent research framework to an international collaborative research system to simultaneously strengthen technological reliability and establish a foundation for global dissemination. In addition, the presence of the following fundamental technological limitations and uncertainties in overseas markets necessitates the adoption of an international collaborative research approach: • Clinically reliable gait and balance data labeling • Algorithm advancement using long-term and diverse user datasets

	• Verification of reliability and reproducibility in real-world usage environments • Design of standardization and commercialization models for global market entry **Why Global Collaboration?** Purdue University, a leading overseas research institution, is a manufacturing-focused engineering university located in Indiana, a region surrounded by numerous manufacturing companies, and serves as an industrial hub in the United States. The university has already been collecting meaningful equipment signal data and IoT sensing data from major manufacturing corporations in the surrounding area and utilizing them for AI-based research. (1) Purdue maintains strong collaborative relationships with medium- to large-scale enterprises equipped with factories, facilities, and production lines; (2) it operates specialized research centers capable of conducting testbed-based research on manufacturing and physical AI; (3) it is strategically located in a major transportation hub in the U.S., making it the only academic institution where Korean companies can directly identify technological trends and market demands in the local environment. Therefore, global collaboration is essential. In addition, Purdue possesses core technologies related to the development of sensor fusion–based data collection pipelines and associated algorithms. As a result, strong synergies with Korean SMEs are expected.
Key Words	1. **Smart Footwear** 2. **Gait & Balance Analysis** 3. **Fall Risk Prediction** 4. **AI-based Digital Health** 5. **Clinical Gait Assessment**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	**R&D Objectives** This project aims to advance a multi-sensor gait data collection system using smart shoes to accurately capture walking and balance patterns. It seeks to establish clinically relevant gait and balance indicators aligned with standardized assessments such as TUG and SPPB, and to develop AI-based algorithms for predicting falls and functional decline. Furthermore, the study focuses on implementing personalized risk prediction models and validating their accuracy and reliability through real-world pilot testing. **Detail R&D Objectives** 1. Advancement of a multi-sensor gait data collection system based on smart shoes 2. Establishment of gait and balance indicator models linked to clinical assessment standards such as TUG and SPPB 3. Development of AI algorithms for predicting falls and functional decline based on left-right asymmetry, COP trajectories, and variability indicators 4. mentation of personalized risk prediction models based on user-specific baseline data 5. Verification of algorithm accuracy and reliability through real-world pilot testing			
R&D Target Specifications	1. Accuracy of TUG test and SPPB physical function assessment: ≥ 90% 2. Fall risk prediction accuracy: ≥ 85% 3. Sensitivity of left-right asymmetry detection: ≥ 90% 4. Error in detecting changes relative to individual baseline: ≤ ±5%			
R&D Plan	**Year 1: Core Technology Development and Enhancement** • Improvement of sensor fusion–based gait data collection algorithms • Mathematical modeling of COP trajectories, asymmetry indicators, and gait variability • Basic performance validation in laboratory environments **Year 2: AI Algorithm Development and Clinical Integration** • Construction of datasets linked to TUG and SPPB assessments • Development of disease and fall risk prediction algorithms • Analysis of repeatability and reproducibility and model optimization **Year 3: Field Validation and Commercialization Linkage** • Pilot testing with real users • Comparative validation between expert evaluations and algorithm results • Technology packaging for service and platform deployment			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Purdue University), led by the overseas partner, will be responsible for designing gait data analysis algorithms and AI models, developing data augmentation and generative AI techniques to address data scarcity, and establishing an analysis framework linked to clinical evaluation. - Design of gait data analysis algorithms and AI models - Development of data augmentation techniques and generative AI to address data scarcity - Establishment of an analysis framework linked to clinical evaluation standards
Expected R&R of Korean SME Partner	The Korean SME partner will take primary responsibility for leading the overall data platform and technology integration, including the development of IoT-based smart shoe hardware, sensor modules, embedded systems, and software. It will also support pilot environment construction and user testing, provide technical feedback from a mass production and commercialization perspective, and collaborate to connect with domestic and international markets. - Overall management of data platforms and technology integration - Development of IoT-based smart shoe hardware and sensor systems - Leadership in the development of sensor modules, embedded systems, and software - Support for pilot environment setup and user testing - Provision of technical feedback from a mass production and commercialization perspective - Collaboration to connect with domestic and international markets

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes** Through this research, clinically reliable AI technologies for gait and balance analysis will be developed, establishing a foundation for the early detection of neuro-musculoskeletal disorders. High-precision analysis algorithms that operate stably across diverse environments and user conditions will be created to enhance applicability in clinical settings. In addition, a data analysis system linked to actual clinical evaluation standards will be established to strengthen the reliability and usability of the technology for medical professionals. **Commercialization Outcomes** Based on the technologies developed in this project, services and platforms will be expanded in the fields of smart healthcare, rehabilitation, and elderly care. By providing personalized health management and remote monitoring services, new added value will be created and market competitiveness will be secured. Furthermore, sustainable business models will be established through cooperation with medical institutions, care facilities, and healthcare companies. **Industrial and Economic Impact** Digital health-based data business models will be created to promote innovation across related industries. The development of new services utilizing collected and analyzed gait and health data will contribute to the expansion of the industrial ecosystem. Through these efforts, the technological

	competitiveness and global market capabilities of small and medium-sized enterprises will be strengthened, contributing to the growth of the national digital health industry.

6. References

(1) Podsiadlo, D., & Richardson, S. (1991). *The Timed "Up & Go": A Test of Basic Functional Mobility for Frail Elderly Persons*.

(2) Guralnik, J. M. et al. (1994). *A Short Physical Performance Battery Assessing Lower Extremity Function*.